Lab Design for Industry 4.0 - Summer Term 2024

Use Case & Requirements

ROVER USED TO CLIMB

Use case:

The primary objective of this project is to design, engineer, and test an autonomous rover capable of climbing steps up to 10 cm in height. The rover must navigate indoor spaces autonomously while preserving stability, maximizing energy efficiency, and ensuring safety.

Requirements:

- **Wheel modification**: Ability to climb steps of 10cm.
- **Rover body**: Lightweight chassis for improved mobility
- **Sensor unit**: LIDAR sensor for obstacle detection.
- **Power source**: Increase battery capacity to ensure continuous operation.
- **Virtual testing:** Conduct virtual testing to climb 10cm stairs.

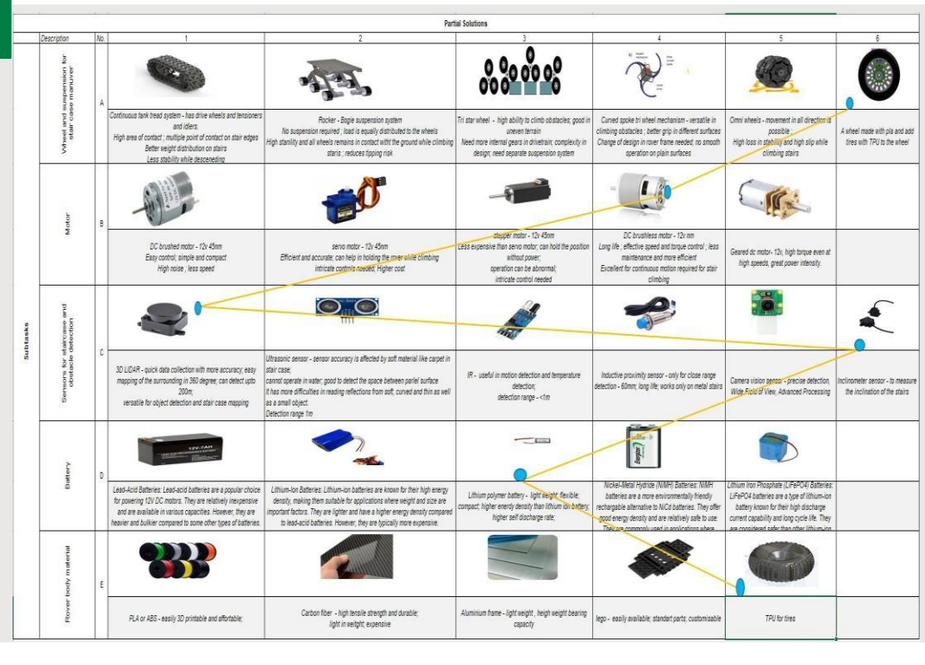
Function structure & Morphological Box

Functional prerequisites:

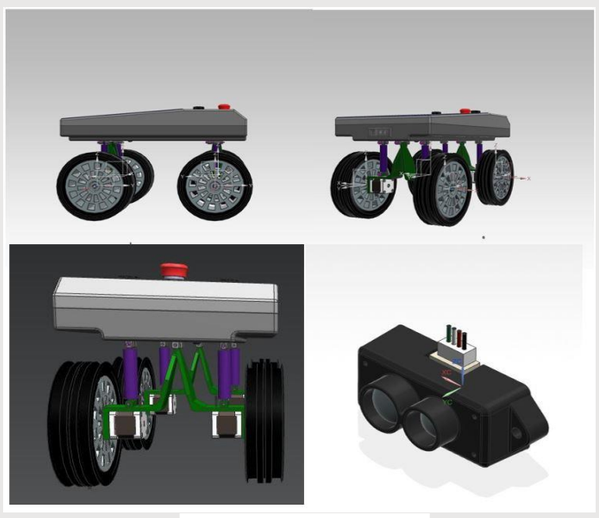
1. **Height Clearance:** The rover must navigate a 10 cm height differential.
2. **Stability & Balance:** Ensure stability while climbing stairs.
3. **Obstacle Detection:** Use sensors to avoid impediments.
4. **Autonomous Operation:** Enable the rover to operate independently during stair climbing.
5. **Reliability:** Ensure the rover climbs stairs without major malfunctions and includes safety features.

Non -functional prerequisites:

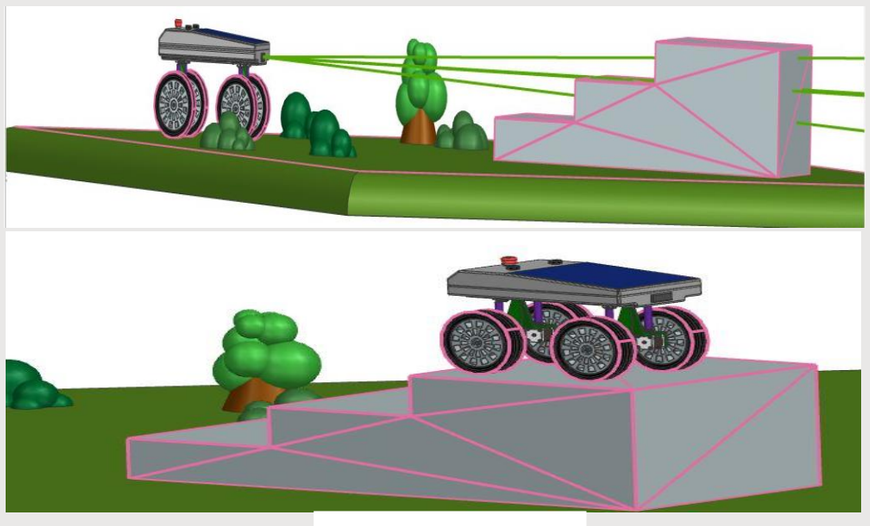
1. **Performance:** The rover should climb stairs smoothly and efficiently, balancing speed and safety.
2. **Power Efficiency:** Maximize battery life to extend operational time between charges.
3. **Environmental Impact:** Ensure the rover operates quietly and without harmful emissions.
4. **Maintenance:** Design for easy maintenance to reduce downtime and repair costs.
5. **Cost-Effectiveness:** Keep the overall cost within budget, considering both initial investment and ongoing operational costs.



CAD Mock-up & Virtual Testing



Rover model



Virtual testing

The models include a virtual commissioning visualization in addition to a regular CAD image.

Conclusion & Reflection

Conclusion:

- **Feasibility Proven**: The successful development and testing of the autonomous rover validated its capability to navigate and climb stairs within indoor environments.
- **Integration of Systems**: The project offered valuable insights into the seamless integration of mechanical systems, control algorithms, and advanced sensor technologies.
- **Enhanced Autonomous Navigation**: The rover's ability to autonomously climb stairs without human intervention was a key achievement, displaying the potential of autonomous robotics in complex environments.

Reflections:

- **Challenges in Optimization**: The project faced significant challenges in optimizing the rover's weight, which was crucial for maintaining stability and ensuring efficient energy consumption.
- **Sensor Accuracy**: Ensuring precise sensor data for navigation and obstacle detection was critical, highlighting the need for high-quality sensors and advanced data processing algorithms.
- **Iterative Design Importance**: The iterative design and testing process proved essential in refining the rover's functionality, leading to improved performance with each iteration.

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